Berea Technical College

Department of Engineering Studies
Mathematics N4 Assignment 2
Subject Code:16030164

Examiner: Mr Y. Mia **Moderator: Mr P. Zvidzai**
Due Date: 27-05-24 **Total: 90**

INSTRUCTIONS

1. **Answer ALL QUESTIONS in full. Show ALL the calculations and intermediary steps. Simplify where possible.**
2. **ALL final answers must be accurately approximated to THREE decimal places.**
3. **Questions may be answered in any order.**
4. **All Formulae used must be written down.**

QUESTION 1 [20]

1.1 Given:

$$6p - q = -3$$
$$2p - q - 4 = 0$$

Use Crammer's rule to solve for p and q. (6)

1.2 Given:

$$\begin{vmatrix} 1 & -3 & 4 \\ -1 & 1 & -5 \\ 2 & 0 & 8 \end{vmatrix}$$

1.2.1 Determine the minor of -3 (2)
1.2.2 Determine the cofactor of 4 (2)

1.3 Given:

$$2I_1 + I_2 - 2I_3 = -1$$
$$3I_1 - 3I_2 - I_3 = 5$$
$$I_1 - 2I_2 + 3I_3 = 6$$

Solve for I_2 only using Crammer's rule. Expand along ROW1 only. (10)

QUESTION 2[16]

2.1 Given: $\frac{x^2}{2} = 4x - 17$ **Solve for** x**:** **(4)**

2.2 Given: $z = \frac{1}{2} - \frac{\sqrt{3}}{2} j$

2.2.1 Convert z **into polar form.** θ **may only be positive.** **(2)**

2.2.2 Indicate z **and all calculated values on the Argand diagram.** **(3)**

2.2.3 Determine $z \times \overline{z}$ **.** **(1)**

2.3 Given: $z = \frac{(1-5j)^2}{1+5j}$

Simplify. Leave the final answer in rectangular form. **(4)**

2.4 Simplify: $(2j)^2 - \frac{1}{j}$ **(2)**

Question 3[20]

3.1 Prove that: $\frac{2\sin^3 x + (\sin 2x)(\cos x)}{2\cos x} = \tan x$ **(5)**

3.2 Simplify the following expression:

$$\frac{\sin(180° + \theta)\cos ec(180° + \theta)\cos(\theta + 360°)}{\cos(90° + \theta)\cos(-\theta)\sec(360 - \theta)}$$ **(5)**

3.3 Solve for θ **:** $5\cos(2\theta) - 3\cos(\theta) + 4 = 0$, $0° \le \theta \le 360°$ **(5)**

3.4 Determine the value of $\frac{1}{2}(\cos 15° + \sqrt{3}\sin 15°)$ **without using a calculator.** **(5)**

Question 4[17]

4.1 Differentiate $y = \sqrt[3]{\cos ec^2 2x}$ **, using a function of a function (chain rule) by using a substitution.** **(4)**

4.2 Differentiate $y = \frac{\ln \cot 3x}{\ln 3x}$ **by using the quotient rule.** **(4)**

4.3 Determine the first derivative of the following with respect to 't' (9)

$$s = \frac{t^4}{2} - 2\ln 5t + coesc4t - \frac{1}{\sec 3t} + 2\pi t^3 - \frac{1}{\sin 4t} + \frac{5^{-3t}}{2} - \frac{1}{4e^{5t}}$$

Question 5[17]

5.1 Evaluate $\int_0^{\frac{\pi}{6}} (3\sin 3x)dx$ **(3)**

5.2 Integrate the following: $\int \left[2e^{-8x} + \frac{\sin 2x}{\sin x} + 3\cos ec^2 6x + \sin\frac{\pi}{2} \right] dx$ **(4)**

5.3.1 Sketch the Area bounded by $y^2 = x$, $y = -x + 1$ and the $y-axis$ **in the first quadrant. Show the representative strip/element (5)**

5.3.2 Calculate the magnitude of the area described in question 5.3.1 (5)

Total: 90